Reonic

Hackathon Challenge · thinc! München 2026

Designing home energy systems in *minutes*, not hours.

Build an AI system that learns from real installer decisions and proposes the right mix of PV, battery storage and heat pump for a home: a design that planners can refine and homeowners can explore.

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00 AT A GLANCE

The Case on One Page

YOUR MISSION

Build an AI system that turns a few facts about a home into a complete energy system design, and that decides the way an experienced planner would.

The problem	Planners at installation companies spend hours designing a system for every single home before an offer can go out. That is time not spent on actual installations.
What you build	From basic project inputs, propose which components a home should get (PV, battery, heat pump, or only some of them) and how large each should be. Users can then adjust the design.
What you get	A large dataset of real residential projects: the input conditions of each home and the system design an installer actually delivered.
Minimum result	For a given project, your system proposes a complete design for the planner with a short explanation, you measure how close your designs come to the expert designs, and the planner can change something and get an updated design.
How we judge	Technical depth, user fit, business impact, and execution and demo. Details on page 8.
Deliverables	A working demo, your code and a short pitch deck. The build starts on **Saturday, 7 November, at 14:00**, submission is on **Sunday, 8 November 2026, at 15:00**.

TERMS USED THROUGHOUT

Installer: a company that sells, plans and installs energy systems. Installers are Reonic's customers.

Planner: the person at an installer who designs systems and writes offers.

Homeowner: the installer's end customer, who receives the offer and lives with the system.

Expert design: the system design an installer actually delivered for a project, as recorded in our data.

AI-Designed Home Energy Systems

Context

Reonic builds the operating system for renewable energy installers. Since 2023, more than 3,000 installation companies in 13 markets across Europe and Latin America have used Reonic to run their business in one place: lead generation, CRM, technical planning, offers and digital signatures, installation and documentation.

This challenge takes you to the step that decides whether a project happens at all: designing the system and turning it into an offer. A homeowner wants lower energy bills and more independence. The installer has to answer one question: **which combination of solar, storage and heat pump fits this specific home?**

The Problem

Designing an energy system for an individual home is one of the most time-consuming steps in the installation process, because every home is different. Annual consumption, household size, the existing heating system, an electric car, the local electricity price and feed-in tariff, and the budget all influence what the right system is.

Planners spend **hours on every single design**. They size the PV system, decide whether a battery pays off and how large it should be, check whether a heat pump makes sense, and balance all of it against what the homeowner can afford. For the installer, that is time not spent where it matters most: on roofs and in basements, installing the systems that have already been sold.

The knowledge to do this well exists, but it lives in the heads of experienced planners. It takes years to learn, it is applied inconsistently, and it does not scale with demand.

Why This Is Technically Hard

The decisions are coupled. A larger PV system changes how valuable a battery is. A heat pump changes the household's consumption and therefore the right PV size. An electric car shifts consumption into the evening. Sizing each component on its own produces systems that look plausible but are wrong.

Expert decisions are not a clean optimisation. Planners apply implicit rules: typical component sizes, budget limits, product availability and homeowner preferences. There is rarely a single correct answer, and real-world data is noisy.

A language model alone will not do it. A strong solution has to reproduce what experts actually decide, stay physically and economically plausible, and explain its recommendations.

The Users

Your solution serves two people whose needs overlap but are not the same. Keep both in mind, and be explicit about where you favour one over the other.

PERSONA 1

The installer's planner

Works at a mid-sized installation company and creates several offers per day, often between site visits. Knows the products well, but is under constant time pressure.

- **Wants:** a solid first design right away that they can trust and build an offer on.
- **Needs control:** can override any part of the design and keeps the final say over the offer.
- **Fears:** a black box that proposes something they cannot justify to the homeowner.

PERSONA 2

The homeowner

Owns a single-family home and has received an offer from an installer for a solar system, possibly with a battery or heat pump. Now has to decide whether to sign.

- **Wants:** to understand the offer and try variants, such as a larger battery or no heat pump, and understand what each change means for them.
- **Needs simplicity:** plain language instead of technical data sheets.
- **Fears:** being oversold, or committing to the wrong system.

The tension: the planner wants speed and full control over a technically and commercially sound offer. The homeowner wants transparency and room to explore, but only within the options the installer allows, for example switching a battery on or off or resizing a component. The home's data stays fixed.

Where to focus: the minimum result serves the planner. A view for the homeowner is a stretch goal that strong teams can add on top of the same engine.

"As Appropriate" Matters

A complete design does not mean every home gets every component. Recommending no battery or no heat pump can be exactly the right expert decision, and your system should be able to make it.

"Every home is different. The expertise to design for it should not be the bottleneck."

02 BUSINESS IMPACT

More Time to Install

The Business Problem

The energy transition is not short of demand. It is short of installer capacity, and design work ties up a large part of it, often for offers that never turn into a signed project.

Slow offers cost deals, because homeowners compare installers and the first credible offer often wins. Inconsistent designs cost trust: an undersized system disappoints, an oversized one loses on price.

Three Use Cases, One Engine

A system that reproduces expert design decisions creates value at three points in the sales process:

PLANNER

Faster offers

The planner enters the basics of a new project and receives a complete design as the starting point for the offer, instead of designing every system from scratch.

PLANNER

Design validation

Before an offer goes out, the planner's own design is compared with what experts delivered for similar homes. Unusual choices, such as an oversized battery, are flagged.

HOMEOWNER

Exploring the offer

A homeowner who has received an offer tries variants of it within the options the installer allows, and understands what each change means before signing.

What This Changes

- **Installer capacity:** less time in the planning tool means more time on site, so the same team completes more projects.
- **Win rate:** faster, consistent and well-explained offers win more often.
- **Quality and trust:** junior planners design like senior ones, and homeowners understand why a system is sized the way it is.

The Broader Opportunity

Every installer who can serve more households accelerates the shift to clean, electrified homes. Turning expert design knowledge into software that scales is a direct lever on how fast that shift happens.

Goals, Data, Demo and Evaluation

Core Building Blocks

Your solution should reach these three goals. How you get there, from model choice to architecture to interface, is entirely up to you.

1. **Propose a complete system.** From a small set of project inputs, decide whether the home gets PV, a battery and a heat pump, and how large each component should be. The proposal must make sense as a whole, not as three independent guesses.

2. **Let the user refine it.** When the planner changes an input or a component, the rest of the design adapts the way an expert would expect, and the planner understands why the system recommends what it does.

3. **Prove it decides like an expert.** Use the data to measure how close your designs come to the expert designs, on projects your system has not seen before.

Ideas to Explore

These questions help you get started. They are not a checklist, and none of them is required.

- What do expert decisions in the data have in common? Are there thresholds or typical bundles that installers apply without writing them down?
- How do you combine learning from past designs with physical and economic reasoning, such as self-consumption, so that results stay plausible for homes unlike anything in the data?
- What is the minimum input you need for a good design, and how do you handle missing or uncertain information?
- How should the system communicate confidence, alternatives and trade-offs?
- What does refinement feel like: sliders, constraints, natural language, or something else?
- How would a validation view tell a planner “this battery is unusually large for a home like this”?
- What can a homeowner change in an offer, and what must stay within the installer’s control?

AI should be a real part of the solution. Statistical and machine learning models trained on the data qualify; a language model is optional. A single prompt to a language model, without grounding in the data, will not reproduce expert decisions reliably.

Expected Outcome

We expect a working slice with an end-to-end demo story, not a production system. Reach the minimum first, then aim higher.

Minimum

- For a given project, your system proposes a complete design, PV, battery and heat pump as appropriate, with a short explanation of why.
- You measure how close your designs come to the expert designs on projects your system has not seen.
- The planner changes at least one input or component and receives an updated design.

Stretch goals

- An analysis of where and why your designs deviate from the experts.
- Alternatives and confidence for every recommendation.
- A validation view for planners, or a view in which homeowners explore their offer, built on the same engine.

The Demo Moment

When we review your submission on Sunday, we want to see a five-minute story like this:

1. **A home is described:** a planner enters a few basic facts about a project.
2. **A complete system appears:** PV, battery and heat pump as appropriate, with a short explanation of why.
3. **The design is refined:** for example, “the family is buying an electric car next year”, and the design adapts sensibly.
4. **It is checked against reality:** the proposal is compared with the expert design for the same project, and you show how well your system does across many projects.

What Reonic Provides

- **A large dataset of real residential projects.** For each project it contains the input conditions, such as energy demand, electricity price, EV usage or an existing heat pump, and the system design that was delivered to the homeowner.
- **Reonic experts on site** throughout the build to answer questions on the data, the domain and how installers work.

Evaluation Criteria

We use these criteria to select our finalists on Sunday. Technology counts just as much as user and business value, so every member of your team can make a difference.

CRITERION	WHAT IT ASKS
Technical depth	How well do your designs reproduce expert decisions, and how convincingly do you measure that across the data?
User fit	Does the solution work for the planner, the homeowner, or both, including refining and understanding a design?
Business impact	Is the value clear, for example in time per offer, design quality or win rate, and would it matter to installers?
Execution and demo	Does it work end to end, and does the demo tell a convincing story?

Submission

Submit the following by **Sunday, 8 November 2026, 15:00**:

- **A working demo** that runs end to end. A short recording as a fallback is recommended.
- **Your code,** for example as a GitHub link, with a short README on setup, approach and known limitations.
- **A short pitch deck** covering the problem, your solution, how you evaluated it and the value for installers and homeowners.

Good to Know

- You are free to choose your own technical approach, models, tools and architecture.
- Do not spend your time on infrastructure. A simple, well-explained solution beats a complex one that does not run.
- Use the data only for this hackathon and do not share it outside your team.
- Stuck on the data or the domain? Ask one of the Reonic experts. That is what we are here for.

We look forward to seeing how you turn expert intuition into software that helps installers build the energy transition faster.